UNSW-NB15 Network Traffic Analysis

Comprehensive Project Architecture, Data Flow, HiveQL Analytics & Runbook

1. Executive Summary & Overview

This project performs distributed Big Data Analytics on network intrusion traces using the modern **UNSW-NB15 dataset**. Built on the **Apache Hadoop** and **Apache Hive** ecosystem, the analytical pipeline leverages distributed SQL-on-Hadoop executed as MapReduce jobs on YARN to process over **2.5 million network connection records**. The investigation targets four critical security dimensions: destination host targeting, flow behavioral profiling, temporal attack patterns, and protocol/service vulnerabilities.

2,540,044	49 Features	9 Attack Categories	Apache Hive 3.1.3
Network Flow Records	Flow & Packet Attributes	Contemporary Threat Types	Distributed SQL on YARN

2. The UNSW-NB15 Dataset & Data Sources

The **UNSW-NB15** dataset was created by the Cyber Range Lab of the Australian Centre for Cyber Security (ACCS) at the University of New South Wales (UNSW Canberra). Using the IXIA PerfectStorm simulation tool, it captures a realistic hybrid of normal enterprise activity and contemporary synthetic cyber-attacks. The 9 attack families represented are: *Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms*.

Column	Inferred Hive Type	Role in Analytics	Used In Query
dstip	STRING	Destination IP address of connection	Query 1
label	INT	Binary target: 0 = Normal, 1 = Attack	Queries 1, 2, 3, 4
attack_cat	STRING	Specific attack category designation	Queries 1, 2
dur	DOUBLE	Record flow duration in seconds	Query 2
sbytes	BIGINT	Source-to-destination transaction bytes	Query 2
dbytes	BIGINT	Destination-to-source transaction bytes	Query 2
stime	BIGINT	Flow start timestamp (Unix epoch seconds)	Query 3
proto	STRING	Transport layer network protocol (TCP/UDP)	Query 4
service	STRING	Application service layer (HTTP, DNS, etc.)	Query 4

3. System Architecture & End-to-End Pipeline

The analytical framework operates through a multi-tiered Big Data pipeline spanning data ingestion, HDFS persistence, schema-on-read modeling in Hive, and parallelized execution on YARN:

Step 1: Raw Ingestion	Download raw UNSW-NB15.csv and feature definitions from remote ACCS repositories.

Step 2: Preprocessing	Clean corrupt records, cast numeric types (sbytes/dbytes to double), format timestamps, prune unneeded columns.
Step 3: HDFS Persistence	Stream clean CSV (UNSW-NB15_prepared.csv) to Hadoop HDFS datanodes via hdfs dfs -put.
Step 4: Hive Schema Definition	Construct external table network_flows_analysis pointing directly to HDFS storage location.
Step 5: Distributed Querying	Execute 4 modular HiveQL queries. Hive parses SQL, generates MapReduce DAGs, and YARN allocates containers.
Step 6: Threat Intelligence	Collate aggregated outputs into tabular metrics summarizing targeted hosts, quantiles, and vulnerability maps.

4. Detailed HiveQL Analysis Modules

Module 1: Destination Address Analysis (destination_address_analysis.sql)

Objective: Identifies the top 15 destination IPs experiencing the highest volume of malicious connections.

Analytical Techniques: Utilizes 3 Common Table Expressions (CTEs) and the ROW_NUMBER() OVER (PARTITION BY dstip ORDER BY category_records DESC) window function. Isolates the single dominant attack family for each destination host along with intra-host attack percentages.

```
WITH destination_summary AS (SELECT dstip, COUNT(*) total_records, SUM(CASE WHEN label=1 THEN 1 ELSE 0 END) attack_records FROM network_flows_analysis GROUP BY dstip)...
```

Module 2: Flow Characteristics Profiling (flow_characteristics.sql)

Objective: Profiles connection duration and data volume (total bytes = sbytes + dbytes) across distinct attack categories.

Analytical Techniques: Computes statistical approximations including mean duration, approximate median (P50), and 95th percentile (P95) using Hive's PERCENTILE_APPROX(col, ARRAY(0.5, 0.95), 10000). Quantifies instantaneous (zero-duration) probe attacks.

```
SELECT attack_cat, ROUND(AVG(dur), 6) mean_dur, ROUND(duration_percentiles[0], 6) approx_median_dur, ROUND(100.0 * zero_duration_records / record_count, 2) zero_duration_pct...
```

Module 3: Hourly Activity Analysis (hourly_activity_query.sql)

Objective: Maps temporal patterns of attack distributions by calendar date and capture hour in UTC.

Analytical Techniques: Enforces hive.local.time.zone=UTC, parses epoch timestamps via FROM_UNIXTIME(stime), and evaluates each hour's contribution to total daily attacks using SUM(attack_records) OVER (PARTITION BY capture_date).

```
SELECT capture_date, capture_hour, attack_records, ROUND(100.0 * attack_records / daily_attack_records, 2) share_of_day_attacks_pct FROM hourly_with_daily...
```

Module 4: Protocol & Service Vulnerabilities (protocol_and_service_query.sql)

Objective: Ranks top 15 transport protocol and application service combinations targeted by malicious flows.

Analytical Techniques: Applies statistical significance thresholding (total_records >= 100) and an unbounded window function SUM(attack_records) OVER () to quantify global attack share for each proto/service group.

```
SELECT proto, service, ROUND(100.0 * attack_records / total_records, 2) within_group_attack_pct,
ROUND(100.0 * attack_records / all_attack_records, 2) share_of_all_attacks_pct...
```

5. Execution Guide & Cluster Specifications

Observed Cluster Specs: Hadoop 3.2.0, Hive 3.1.3 (YARN engine), OpenJDK 1.8.0_502, Scala 2.12.18, Spark 3.5.9 (pending), 31 GiB RAM + 8 GiB Swap, ~51.25 GB HDFS storage.

6. Readiness Assessment & Critical Blockers

Component	Status	Assessment & Action Required
HiveQL Analysis Scripts	COMPLETE	All 4 SQL queries are complete, robust, and production-ready.
Documentation (README)	COMPLETE	Thorough specifications of environment, dependencies, and dataset citations.
Data Preparation Script	MISSING	**Blocker:** Transformation script to produce UNSW-NB15_prepared.csv is omitted.
Hive DDL Table Script	EMPTY	**Blocker:** hive_table_definitions.sql is 0 bytes; external table must be defined manually.
Raw Dataset (CSV)	EXTERNAL	Gitignored (correct practice); must be downloaded via external Dropbox link.
Apache Spark Approach	PENDING	Roadmap item documented in README; directory and scripts not yet implemented.